

A
SYNOPSIS REPORT
ON
CityWatch

Submitted in partial fulfilment of the requirements of the degree of
MCA Department of Trinity Academy of Engineering



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Project Based Learning

Title of Project	City Watch
Operating System	Windows 11
Processor	Intel core i7
Front End	Angular,flutter
Back End	springboot
Database	MySQL

1 About Project

1.1 Problem Statement

In rapidly growing urban areas, citizens often face difficulties in reporting civic issues such as road damage, sanitation problems, water supply disruptions, streetlight failures, and public safety concerns. Traditional complaint mechanisms, which rely on manual processes or disconnected communication channels, lead to delays, lack of transparency, poor tracking, and inefficient resolution of public grievances.

There is no centralized digital platform that allows citizens to register complaints, monitor their status, and communicate effectively with local authorities based on their ward or locality.

Additionally, the absence of secure authentication systems and proper complaint categorization further complicates accountability and response management.

Therefore, there is a need to develop a secure, scalable, and user-friendly web-based system that enables citizens to report issues, track complaint progress, and receive updates while allowing authorities to manage and resolve complaints efficiently.

The proposed system, CityWatch – Community Safety and Complaint Management System, aims to bridge this gap by providing a structured, transparent, and technology-driven solution using modern web technologies such as Angular, Spring Boot, and MySQL

1.2 Objective

The main objectives of the CityWatch – Community Safety and Complaint Management System are as follows:

- **Centralized Complaint Management:**

Provide citizens with a secure and centralized digital platform to register and track civic complaints related to infrastructure, sanitation, safety, and public services.

- **Real-Time Complaint Tracking:**

Enable users to monitor the status of their complaints and receive timely updates from authorities.

- **Ward-Based Categorization:**
Organize complaints based on ward number and location to ensure efficient handling and faster resolution.
- **Secure Authentication System:**
Implement a robust authentication mechanism including email verification and OTP-based password reset to ensure user security.
- **Admin Dashboard for Authorities:**
Provide local authorities and administrators with a dashboard to manage, update, and resolve complaints efficiently.
- **User-Friendly Interface:**
Design an intuitive and responsive web interface using Angular and Tailwind CSS for seamless navigation and accessibility.
- **Scalability and Future Enhancements:**
Develop the system using scalable technologies such as Spring Boot and MySQL to allow future integration of mobile applications, geo-location tagging, and real-time notifications.

1.3 Scope of System

The CityWatch – Community Safety and Complaint Management System aims to provide a centralized and transparent digital platform for citizens and local authorities to manage civic issues efficiently. The scope of the system includes:

- **User Base:**
 - **Citizens:** To register, log in, submit complaints, and track the status of issues related to their locality.
 - **Administrators / Authorities:** To monitor, categorize, manage, and resolve complaints efficiently at ward and city levels.
- **Core Functionalities:**
 - **User Registration and Authentication:** Secure account creation, login, and OTP-based password recovery.
 - **Complaint Management:** Allow users to submit complaints with details such as ward number, city, and description.
 - **Ward-Based Categorization:** Organize complaints according to ward numbers for structured management.
 - **Complaint Status Tracking:** Enable users to view real-time updates on complaint progress.
 - **Admin Dashboard:** Provide authorities with tools to update status, respond to complaints, and manage records.
- **Technology and Deployment:**
 - **Frontend Development:** Web-based user interface developed using Angular and Tailwind CSS.
 - **Backend Services:** Developed using Spring Boot for secure and efficient server-side processing.
 - **Database Management:** MySQL database for structured, reliable, and scalable data storage.

- **Security Mechanisms:** Implementation of JWT authentication and OTP-based email verification.
- **Future Enhancements:**
 - **Mobile Application Integration:** Development of an Android/iOS mobile version of the system.
 - **Geo-Location Tagging:** Enable users to attach location data to complaints for accurate issue reporting.
 - **Real-Time Notifications:** Integration of SMS or push notifications for status updates.
 - **Data Analytics Dashboard:** Provide graphical insights and reports for better decision-making.

1.4 Methodology

- **Requirement Analysis:**
 - Identify system functionalities such as user registration, secure login, complaint submission, complaint tracking, and admin management.
 - Analyze ward-based categorization requirements and complaint status workflow.
 - Select appropriate technologies for frontend, backend, database, and security implementation.
- **Database Design:**
 - Design relational tables for Users, Complaints, Wards, and Password Reset Tokens.
 - Establish relationships between users and complaints for structured data handling.
 - Implement constraints and validation rules to maintain data integrity and security.
- **Backend Development:**
 - Develop RESTful APIs using Spring Boot for user management and complaint operations.
 - Implement business logic for complaint registration, status updates, and ward-wise filtering.
 - Integrate secure authentication using JWT and OTP-based password reset via email.
- **Frontend Development:**
 - Build a responsive and user-friendly web interface using Angular and Tailwind CSS.
 - Implement reactive forms with validation for registration, login, and complaint submission.
 - Connect frontend services to backend APIs for real-time data interaction.
- **Security Implementation:**
 - Implement JWT-based authentication and role-based authorization.
 - Integrate OTP-based email verification for secure password recovery.
 - Ensure proper validation and sanitization of user inputs.
- **Testing and Evaluation:**
 - Perform unit testing for backend APIs and database operations.

- Conduct integration testing between Angular frontend and Spring Boot backend.
- Carry out usability testing to ensure smooth user experience.

- **Deployment:**

- Deploy the backend on a secure server environment.
- Configure database hosting for reliability and scalability.
- Monitor application performance and optimize system efficiency.

References

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- [4] Jones, M., Bradley, J., Sakimura, N. (2015). JSON Web Token (JWT). RFC 7519. Internet Engineering Task Force (IETF).
- [5] Heeks, R. (2006). Implementing and Managing eGovernment: An International Text. SAGE Publications.